MULUNGUSHI UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

ICT – 202 DATA STRUCTURES AND ALGORITHMS II

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PROGRAMME: COMPUTER SCIENCE

TITLE: CONVERTING INFIX EXPRESSIONS TO POSTFIX AND PREFIX

DATE: 14th April 2026

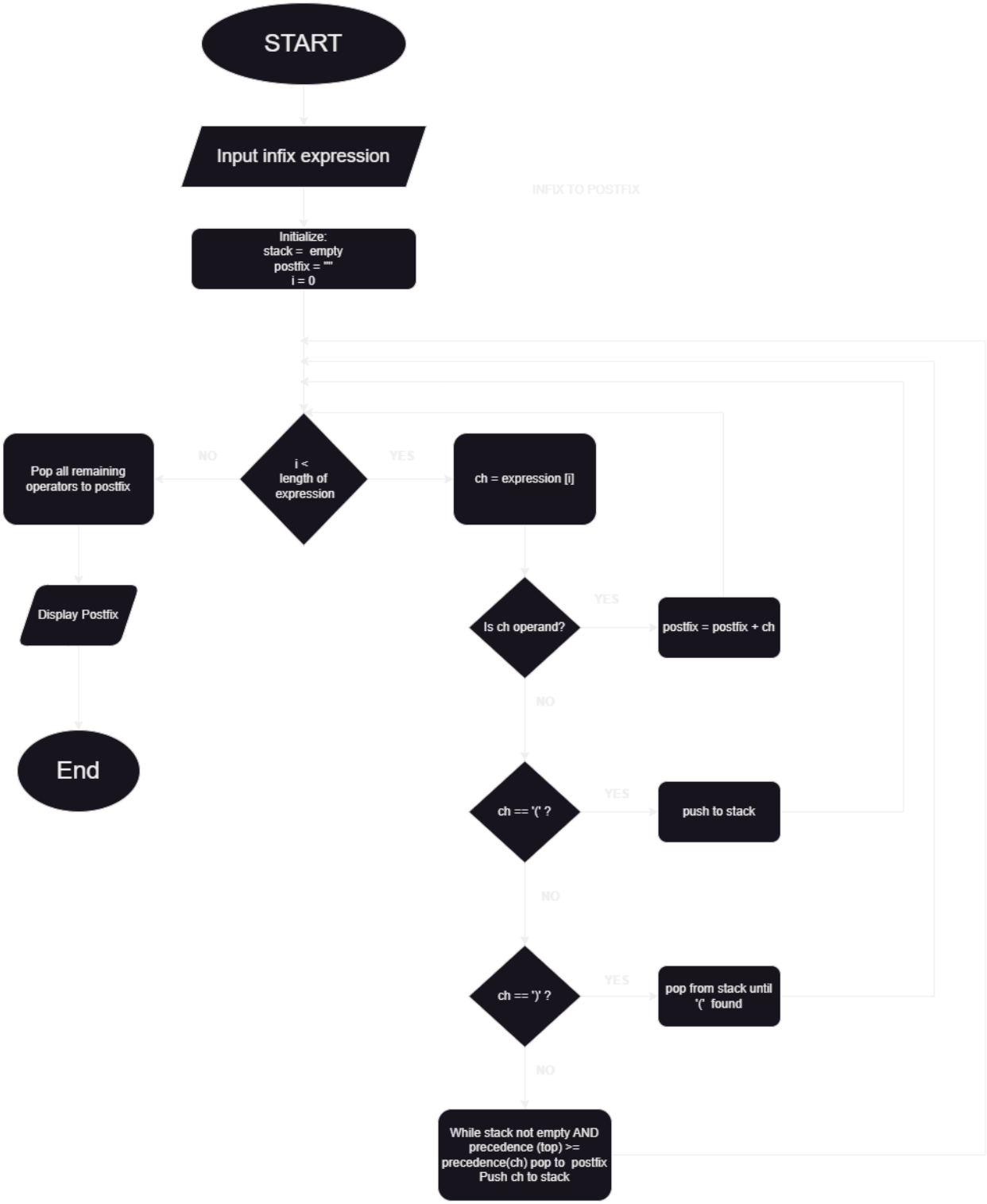
DUE DATE: 16th April 2026

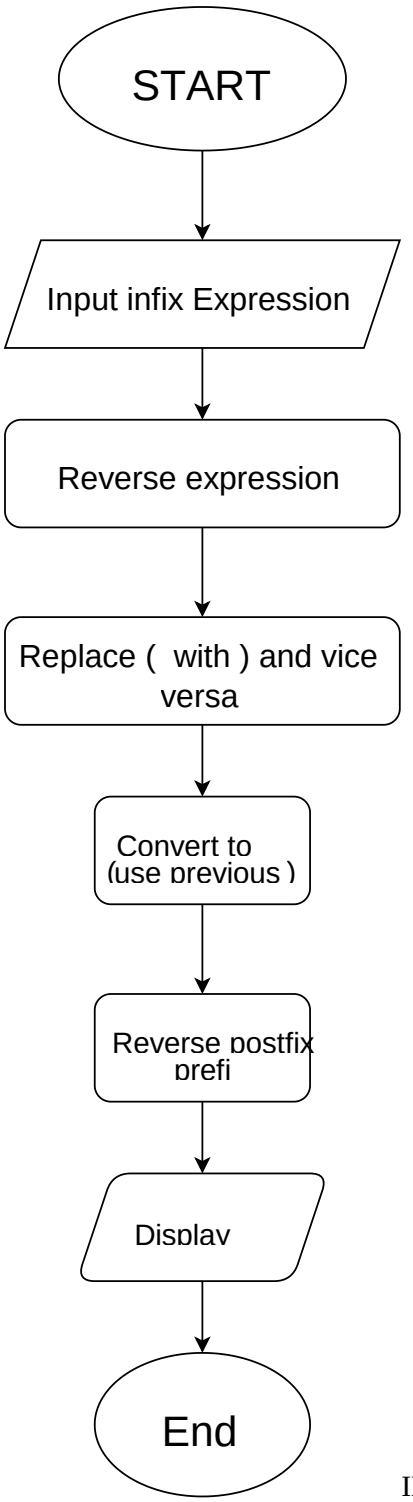
LECTURER: KALUBA Z. (MR.)

PSEUDOCODE

```
FUNCTION infixToPostfix(expression)
CREATE empty stack
CREATE empty postfix

FOR each character ch

        IF operand → add to postfix
        IF '(' → push to stack
        IF ')' → pop until '('
        ELSE operator:
                WHILE precedence (top) >= precedence (ch)
                        pop to postfix
                push ch
END FOR

POP remaining stack to postfix
RETURN postfix
END FUNCTION

FUNCTION infixToPrefix(expression)
REVERSE expression
SWAP brackets
CONVERT to postfix
REVERSE result
RETURN prefix
END FUNCTION
```

WORKFLOW


START
Input infix expression
INFIX TO POSTFIX
Initialize:
stack = empty
postfix = ""
i = 0
i <
length of
expression
NO
YES
Pop all remaining
operators to postfix
Display Postfix
End
ch = expression [i]
Is ch operand?
YES
postfix = postfix + ch
NO
ch == '(' ?
YES
push to stack
NO
ch == ')' ?
YES
pop from stack until
'(' found
NO
While stack not empty AND
precedence (top) >=
precedence(ch) pop to postfix
Push ch to stack



INFIX TO PREFIX